

Pediatric Brain Tumor Segmentation via nnU-Net with 3D Printing Validation

Tarunpreet Kaur¹, Vivek Jain², Pratham Handa¹, Gourav Jain³, and Nimrat Kaur⁴

¹ Department of Computer Science and Engineering, Thapar Institute of Engineering and Technology

² Department of Mechanical Engineering, Thapar Institute of Engineering and Technology

³ Department of Electronics and Communication Engineering, Thapar Institute of Engineering and Technology

⁴ Department of Biotechnology, Thapar Institute of Engineering and Technology

Abstract. Pediatric brain tumors represent one of the most challenging diagnoses in pediatric oncology, requiring accurate segmentation for treatment planning and monitoring. These tumors exhibit unique imaging characteristics, including small enhancing components and high anatomical variability, as demonstrated in the BraTS-PED 2025 challenge dataset comprising 438 multi-parametric MRI cases (260 training, 91 validation, 87 test). While manual annotation is time-consuming, subject to inter-observer variability, and limited by the availability of specialized radiologists, deep learning-based automated segmentation offers an optimized alternative. The proposed nnU-Net v2 framework employs pediatric-specific preprocessing with brain-masked z-score normalization, extensive data augmentation, automated intensity-based label mapping, and connected-component filtering. The method achieves test set Dice scores of 0.882 ± 0.103 (whole tumor), 0.542 ± 0.380 (tumor core), and 0.590 ± 0.374 (enhancing tumor), with Hausdorff distances of 6.73 mm (WT), 59.07 mm (TC), and 66.67 mm (ET). The framework extracts tumor parameters including volumes (cm^3) for all regions, maximum cross-sectional areas with slice indices, centroids in voxel and millimeter coordinates, bounding box extents, and connected component counts, providing quantitative biomarkers for treatment planning and monitoring. To enhance clinical interpretability and surgical planning, segmented tumors were reconstructed into patient-specific 3D surface models and fabricated using 3D printing. Volumetric discrepancies between digital and printed models were 3–5%, and linear dimensional errors were below 2 mm, confirming geometric fidelity.

Keywords: Pediatric Brain Tumors · Semantic Segmentation · nnU-Net · Deep Learning · Tumor Biomarkers

1 Introduction

Pediatric brain tumors present fundamentally different challenges compared to adult gliomas in medical imaging, requiring accurate segmentation to guide sur-

gical planning, radiation therapy targeting, and treatment monitoring. Unlike adult cases, pediatric tumors exhibit smaller and often absent enhancing components (present in only 73.8% of cases), higher inter-patient anatomical variability, distinct contrast-enhancement patterns, and inconsistent annotation protocols across clinical datasets [1, 2]. These unique characteristics make pediatric tumor segmentation particularly challenging and prone to class imbalance and label noise.

Multi-parametric magnetic resonance imaging (mpMRI) is the standard non-invasive modality for pediatric brain tumor assessment, typically comprising T1-weighted (T1), contrast-enhanced T1-weighted (T1ce), T2-weighted (T2), and T2-weighted fluid-attenuated inversion recovery (FLAIR) sequences [3]. Manual segmentation by expert radiologists is time-intensive, subject to inter-observer variability, and dependent on specialist expertise. These challenges are particularly pronounced in pediatric neuro-oncology, where tumor presentations are heterogeneous and less common [3].

Recent advances in deep learning have demonstrated exceptional capabilities in automating medical image segmentation tasks. The nnU-Net framework [4] has emerged as an efficient baseline for biomedical image segmentation, automatically configuring preprocessing pipelines, network architectures, and training procedures for new tasks. However, pediatric applications face distinct obstacles: enhancing tumor components are frequently very small (<1,000 voxels) or absent, label assignments vary across institutions with different numerical indices for identical tissue classes, and standard adult preprocessing strategies do not transfer directly to pediatric populations. This work addresses these challenges through preprocessing adapted for pediatric MRI, automatic label standardization based on image intensities (avoiding manual re-annotation), and anatomy-aware post-processing to improve prediction quality, achieving competitive performance on BraTS-PED 2025 with efficient training (16 hours per fold) and inference (20 seconds per case).

2 Related Work

Deep learning approaches to brain tumor segmentation have evolved from manually engineered U-Net variants toward automated and task-adaptive frameworks. Raza et al. developed GARU-Net, incorporating gated attention mechanisms and GELU activations, achieving Dice scores of 0.860–0.908 on BraTS 2020 adult glioma data [5]. The introduction of nnU-Net [4] marked a shift from architecture-centric innovation to automated configuration, where preprocessing, network design, and training schedules are adapted to dataset characteristics, establishing it as the standard baseline across BraTS challenges.

Building upon this foundation, recent BraTS-PED 2025 studies have emphasized pediatric-specific optimization and ensemble approaches. Yi et al. demonstrated that integrating nnU-Net, Swin UNETR, and frequency-aware decomposition (HFF-Net) improves segmentation of small subregions, achieving Dice scores of 72.3% (ET), 95.6% (NET), and 92.3% (WT) [6]. Beyond model back-

bone design, Capellán-Martín et al. showed that radiomic-guided, lesion-aware ensemble strategies with adaptive cross-validation and structured post-processing can yield competitive performance across BraTS 2025 tasks without reliance on a single architecture [7]. Similarly, Li et al. proposed targeted nnU-Net refinements, including residual encoders with attention mechanisms and specificity-driven regularization, achieving top validation performance on BraTS-PED 2025 and highlighting the importance of pediatric-focused architectural adjustments [8].

Kazerooni et al. established the BraTS-PED challenge [1, 2], assembling the largest annotated pediatric brain tumor dataset to date (438 cases) with systematic inter- and intra-rater variability analysis. Their findings underscored key pediatric-specific challenges: enhancing tumor (ET) regions are substantially smaller than in adults, with 26.2% of cases containing no enhancing tissue, and label index inconsistencies across institutions complicate supervised training. Furthermore, standard preprocessing strategies developed for adult gliomas often fail to generalize due to variable skull-stripping quality and extreme class imbalance, with ET frequently representing less than 0.1% of total voxels.

3 Methods

An overview of the complete pipeline is shown in Fig. 1, comprising multi-parametric MRI input through to quantitative tumor parameter extraction.

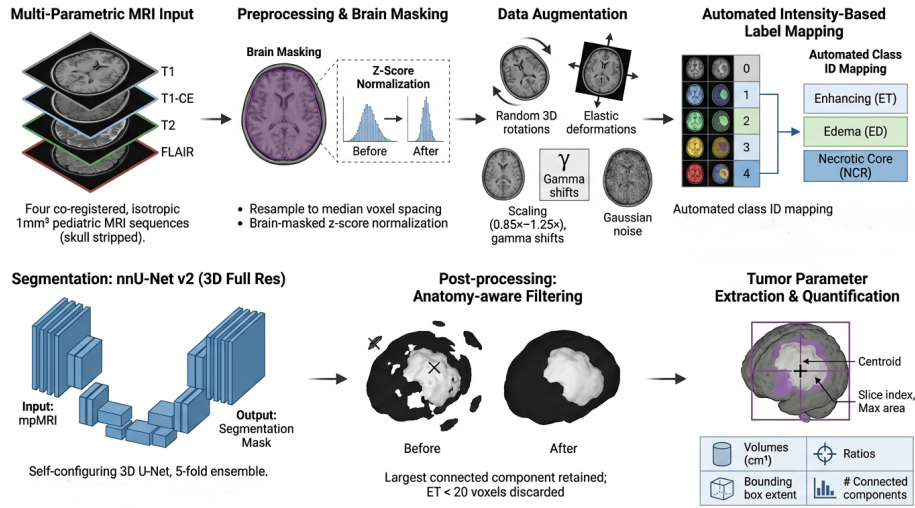


Fig. 1. Overview of the brain segmentation pipeline comprising multi-parametric MRI input to quantitative tumor parameter extraction.

3.1 Dataset

The BraTS-PED 2025 dataset contains 438 multi-parametric MRI scans from pediatric patients with brain tumors, distributed across 260 training cases, 91 validation cases, and 87 test cases. Each case comprises four co-registered MRI sequences: T1-weighted (T1), contrast-enhanced T1-weighted (T1ce), T2-weighted (T2), and T2-weighted Fluid Attenuated Inversion Recovery (T2-FLAIR). All images are skull-stripped and resampled to $1.0 \times 1.0 \times 1.0$ mm isotropic spacing.

Manual annotations delineate tumor subregions using labels $\{0, 1, 2, 3, 4\}$, where 0 represents background and 1–4 represent tumor tissue classes. However, the semantic assignment of these numerical labels to specific tissue types (enhancing tumor, necrotic core, edema) was not standardized, requiring automated inference. The dataset exhibits substantial biological heterogeneity: tumor volumes range from 0.5 cm^3 to $400+ \text{ cm}^3$, enhancing components are absent in 26.2% of cases, and tumor locations span all brain lobes with varying proximity to critical structures.

3.2 Preprocessing and Augmentation

All imaging volumes undergo standardized preprocessing before network training. First, modalities are resampled to a common median voxel spacing to handle resolution heterogeneity typical in pediatric acquisitions. Intensity normalization is performed per-modality per-subject using z-score standardization:

$$\hat{x} = \frac{x - \mu_{\text{mask}}}{\sigma_{\text{mask}} + \varepsilon} \quad (1)$$

where μ_{mask} and σ_{mask} are computed strictly within the non-zero brain mask region to avoid bias from background voxels. This conservative brain mask is created by combining all non-zero voxels from the four modalities. To prevent overfitting on the pediatric dataset (260 training cases), on-the-fly data augmentation is applied during training using the `batchgenerators` framework [9]. Augmentation includes random 3D rotations, elastic deformations, anisotropic scaling ($0.85\times$ to $1.25\times$), gamma intensity changes, and Gaussian noise, applied randomly during each training epoch.

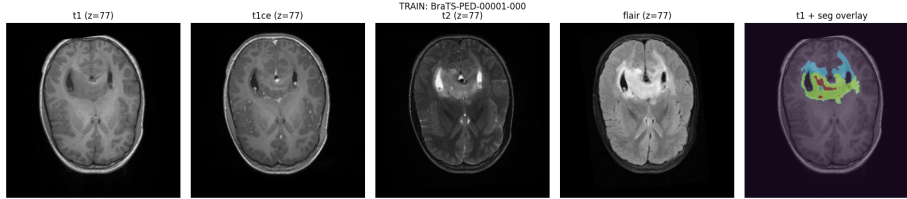


Fig. 2. Multi-parametric MRI input showing T1, T1ce, T2, FLAIR modalities and ground truth segmentation for case BraTS-PED-00001-000.

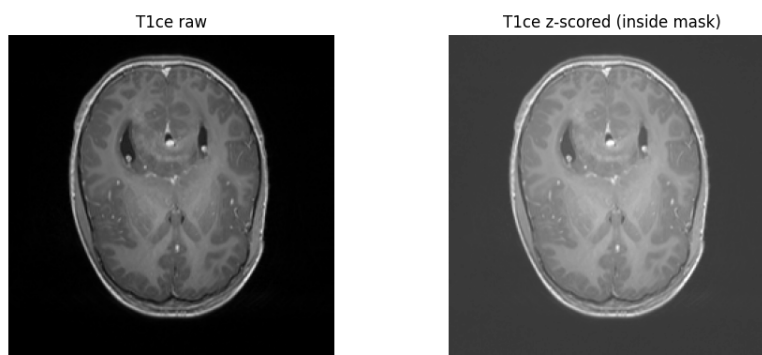


Fig. 3. Effect of z-score normalization on T1ce modality, showing intensity distribution before (left) and after (right) normalization within the brain mask.

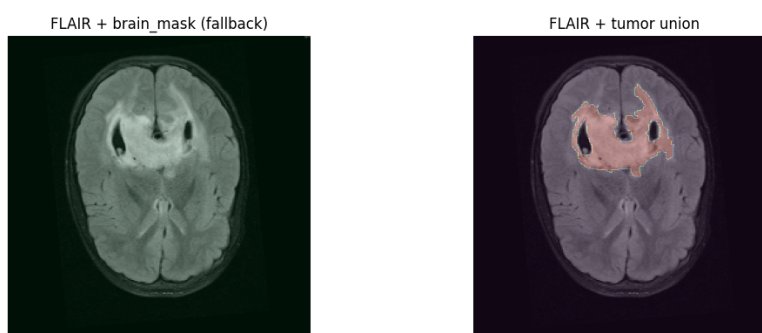


Fig. 4. Brain mask generation (left) and whole tumor region (right) used for intensity normalization.

3.3 Intensity-Based Label Mapping

The dataset contained labels $\{0, 1, 2, 3, 4\}$ with ambiguous tissue assignments that varied across contributing institutions. To identify the numerical label corresponding to each tissue type without manual expert re-annotation, an automated mapping procedure was developed that exploits the distinctive MRI intensity patterns of different tumor tissues.

The procedure samples $N = 20$ training volumes and computes the mean intensity for each label after within-brain z-score normalization. The key discriminative feature is the contrast enhancement signal, calculated as the difference between T1ce and T1 intensities:

$$\Delta_{\text{CE}}(\ell) = Z_{T1ce}(\ell) - Z_{T1}(\ell) \quad (2)$$

where $Z_{\text{mod}}(\ell)$ denotes the mean z-scored intensity of label ℓ in the given modality. Enhancing tissue exhibits a strong positive Δ_{CE} due to gadolinium uptake on contrast-enhanced images, whereas necrotic tissue shows low intensity in both sequences. Based on these intensity signatures combined with volume priors, labels are categorized:

- **Enhancing Tumor (ET):** Label with highest median Δ_{CE} and smaller volume (original label 1, $\Delta_{\text{CE}} = 0.508$).
- **Peritumoral Edema (ED):** Label with the highest FLAIR intensity and larger volume (original label 4, $Z_{\text{FLAIR}} = 2.588$).
- **Necrotic Core (NCR):** Remaining tumor labels with low contrast enhancement (original labels 2 and 3 combined).

The inferred mapping was $\{0 \rightarrow 0, 1 \rightarrow \text{ET}(4), 4 \rightarrow \text{ED}(2), \{2, 3\} \rightarrow \text{NCR}(1)\}$, remapping original dataset labels to BraTS-standard label indices $\{0, 1, 2, 4\}$ where $\text{ET} = 4$, $\text{ED} = 2$, and $\text{NCR} = 1$. As nnU-Net v2 requires consecutive integer labels, a final conversion of $\text{ET } 4 \rightarrow 3$ was applied before training, yielding the internal label set $\{0, 1, 2, 3\}$; all evaluation metrics are reported in the original BraTS convention ($\text{ET} = 4$, $\text{TC} \in \{1, 4\}$). Because label IDs may vary across contributing institutions, the mapping is derived from MRI intensity signatures rather than assuming fixed label meanings. The discriminative patterns are illustrated in Fig. 5.

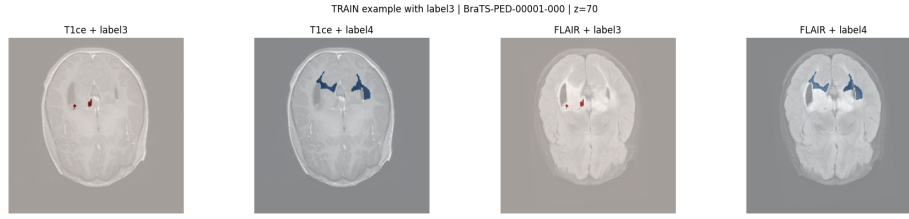


Fig. 5. Intensity-based label discrimination showing T1ce and FLAIR patterns for label 1 (enhancing, red) versus labels 2 and 3 (necrotic, blue) used in automated label mapping.

3.4 Network Architecture and Training

The nnU-Net v2 framework automatically configures a 3D U-Net architecture based on dataset properties, including patch size, batch size, and network depth [4]. The model uses a Dice-cross-entropy combined loss function and is trained for 1,000 epochs with a batch size of 2. The `3d_fullres` configuration was trained on an NVIDIA H100 GPU; training completed in approximately 16 hours per fold, verified from training log timestamps. Predictions from the five cross-validation folds were used to produce the final results. Where fold outputs overlap, majority voting is applied; otherwise, the predictions are aggregated over the union of cases.

3.5 Post-Processing

An optional post-processing step of connected-component filtering is applied to remove anatomically implausible regions. Two operations are performed in sequence. The whole tumor mask $M_{WT} = \mathbf{1}[y > 0]$ is first decomposed into 3D connected components using 26-connectivity, retaining only the largest component. This eliminates spatially disconnected false-positive islands caused by skull-stripping residuals or intensity artifacts. Small isolated regions of predicted enhancing tumor ($M_{ET} = \mathbf{1}[y = 3]$) containing fewer than $\tau = 20$ voxels are discarded and relabeled as background. Post-processing was applied to all test set predictions prior to submission.

3.6 Tumor Parameter Extraction

Beyond segmentation masks, the framework extracts clinically relevant tumor biomarkers for treatment planning and monitoring. Volumes are computed from voxel counts and NIfTI header spacing information, converted to cm^3 ($1 \text{ cm}^3 = 1,000 \text{ mm}^3$). The following parameters are extracted:

- **Volumes (cm^3):** WT, TC, ET, ED, NCR, and NET computed from voxel counts $\times$ voxel volume.

- **Volume Ratios:** ET/WT and TC/WT ratios for tumor composition assessment.
- **Maximum Cross-Sectional Area:** Largest axial slice area (cm^2) and corresponding slice index for WT.
- **Centroids:** Tumor center-of-mass in both voxel coordinates and millimeter coordinates.
- **Bounding Box Extent:** 3D extent in millimeters (width $\times$ height $\times$ depth) for radiation field planning.
- **Connected Component Counts:** Number of spatially disconnected tumor regions for WT and ET.

3.7 Evaluation Metrics

Segmentation performance is evaluated using the Dice similarity coefficient (DSC) and 95th percentile Hausdorff distance (HD95) for three composite regions following BraTS conventions

:

- **Whole Tumor (WT):** All tumor tissue and edema ($y > 0$).
- **Tumor Core (TC):** Necrotic core and enhancing tumor ($y \in \{1, 3\}$, corresponding to BraTS labels NCR = 1 and ET = 4 prior to nnU-Net conversion).
- **Enhancing Tumor (ET):** Enhancing components only ($y = 3$, corresponding to BraTS label ET = 4 prior to nnU-Net conversion).

The Dice coefficient measures volumetric overlap between prediction and ground truth. HD95 quantifies boundary distance errors, particularly relevant for radiation therapy planning, where dose margins must account for segmentation uncertainty. When both prediction and ground truth are empty for a region, Dice is defined as 1.0; when exactly one is empty, Dice is 0.0.

4 Results

4.1 Segmentation Performance

The nnU-Net v2 `3d_fullres` model converged after 1,000 epochs. Table 1 presents Dice scores and Hausdorff distances for training and test sets, demonstrating strong whole tumor segmentation with more challenging subregion performance.

Whole tumor segmentation achieved research-grade quality (Dice > 0.80) on both evaluations. The performance gap between training and test sets for TC ($0.800 \rightarrow 0.542$) and ET ($0.805 \rightarrow 0.590$) reflects biological heterogeneity in pediatric tumors with gradual, ill-defined subregion boundaries. HD95 metrics confirm spatial accuracy suitable for radiation therapy planning: whole tumor boundaries are accurate to within 6.73 mm, while tumor core and enhancing tumor show larger boundary uncertainties (59.07 mm and 66.67 mm, respectively) due to smaller region sizes and lower contrast. Inference averaged 20 seconds per case on test volumes, demonstrating clinical deployment feasibility.

Table 1. Segmentation performance on training and test sets showing Dice scores and 95th percentile Hausdorff distances (HD95) in millimeters.

Evaluation	WT Dice	TC Dice	ET Dice	Mean Dice	Cases
Training Set	0.950	0.800	0.805	0.852	260
Test Set (Dice)	0.882	0.542	0.590	0.671	30
Test Set (HD95, mm)	6.73	59.07	66.67	44.16	—

4.2 Tumor Parameter Analysis

A representative segmentation alongside extracted tumor parameters is shown in Fig. 6 for case BraTS-PED-00001-000. The nnU-Net prediction yields: WT volume = 83.70 cm^3 , TC volume = 13.83 cm^3 , ET volume = 6.84 cm^3 , and ED volume = 69.87 cm^3 . Volume ratios are $\text{ET/WT} = 0.082$ and $\text{TC/WT} = 0.165$, indicating a predominantly edematous tumor with minimal enhancing components (10 connected ET components). The maximum axial WT cross-sectional area is $2,672.0 \text{ mm}^2$ at slice index $z = 96$. The WT bounding box extent is $67.0 \times 81.0 \times 59.0 \text{ mm}$, providing radiation field dimensions. These quantitative biomarkers enable objective assessment of tumor burden, treatment response monitoring through longitudinal volume changes, and automated radiation therapy field planning based on bounding box extents.

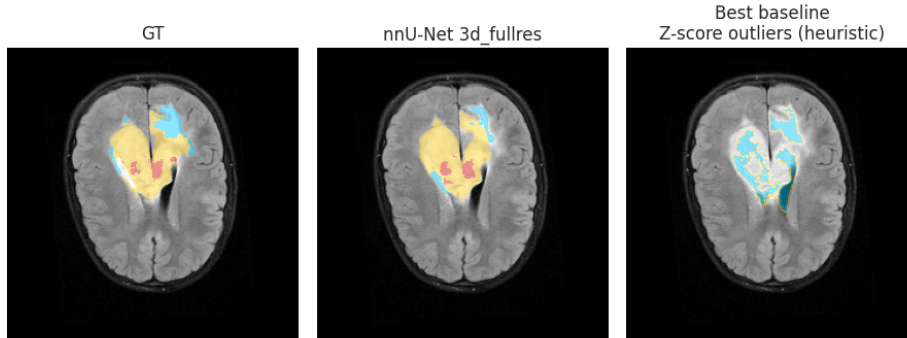


Fig. 6. Segmentation comparison showing ground truth and nnU-Net prediction with extracted tumor parameters for case BraTS-PED-00001-000.

4.3 3D Model Reconstruction and Physical Fabrication

To enhance clinical interpretability, tumor subregion masks were reconstructed into patient-specific physical models using an end-to-end 3D printing pipeline. Binary segmentation masks for WT, TC, and ET subregions were exported as

NIfTI volumes and converted into triangular surface meshes using the Marching Cubes algorithm [11], then saved as STL files compatible with standard 3D printing toolchains.

Raw meshes contained staircase artifacts from voxel discretization and were post-processed in MeshLab [12] and Blender [13] using Laplacian smoothing, hole filling, and mesh repair to produce watertight geometry. The three sub-region meshes were retained as separate objects to enable colour-differentiated visualization of tumor composition. Scaling was maintained at 1:1 anatomical scale, with print orientation optimized for structural integrity. STL models were sliced in Bambu Studio and fabricated using Fused Deposition Modelling (FDM) with PLA filament (layer height 0.1–0.2 mm, nozzle diameter 0.4 mm, infill 20%), a well-established approach for medical anatomical prototyping [14, 15].

Geometric fidelity was validated by comparing digital and printed models volumetrically and linearly. Reference volumes were derived from NIfTI voxel spacing metadata; printed volumes were measured by water displacement and cross-validated with digital calipers at standardized landmarks including maximum tumor diameter, bounding box extents, and principal cross-sectional widths. Volumetric discrepancies were 3–5% and linear errors were below 2 mm, consistent with FDM tolerances reported for neurosurgical planning models [14, 15]. The resulting physical models support pre-operative visualization, surgical rehearsal, and clinician–patient communication.

5 Discussion

The framework was designed around the specific properties of pediatric MRI rather than adapted from adult glioma pipelines. Brain-masked z-score normalization excludes background voxels that would skew intensity statistics under variable skull-stripping quality. The intensity-based label mapping exploits the known biophysical contrast between gadolinium-enhancing and necrotic tissue to resolve inconsistent label assignments across institutions, eliminating the need for manual re-annotation. nnU-Net v2 was selected for its automatic adaptation to class imbalance, which is critical given that enhancing tumors can represent less than 0.1% of total voxels.

Whole tumor segmentation (Dice 0.882, HD95 6.73 mm) falls within the 5–10 mm planning margins used in pediatric radiotherapy, confirming clinical usability. Lower performance on tumor core and enhancing tumor (Dice 0.542 and 0.590) reflects the small spatial extent of these subregions, ill-defined boundaries, and complete absence of enhancing tissue in 26.7% of cases. The single-model approach prioritizes deployability and transparency, with approximately 16-hour training per fold and 20-second inference, offering a practical path to clinical integration. Physical validation via 3D-printed models confirmed geometric fidelity, with volumetric error 3–5% and linear error below 2 mm. Future work will focus on uncertainty estimation to flag low-confidence predictions for radiologist review.

6 Conclusion

The proposed study presents an automated framework for pediatric brain tumor segmentation that addresses dataset-specific challenges through intensity-based label mapping, pediatric-adapted preprocessing, and comprehensive tumor parameter extraction. The study achieves competitive performance on BraTS-PED 2025 (WT Dice 0.882, TC Dice 0.542, ET Dice 0.590) with efficient training (approximately 16 hours per fold) and inference (20 seconds per case) suitable for clinical deployment. The intensity-based mapping procedure provides a practical solution for handling ambiguous label assignments in multi-institutional datasets without requiring manual re-annotation. Extracted tumor parameters, including volumes, ratios, centroids, bounding boxes, and component counts, provide quantitative biomarkers for treatment planning, therapy field design, and longitudinal response monitoring. Future work should focus on improving subregion segmentation through advanced architectures, implementing uncertainty estimation for clinical decision support, and validating on independent multi-institutional datasets.

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